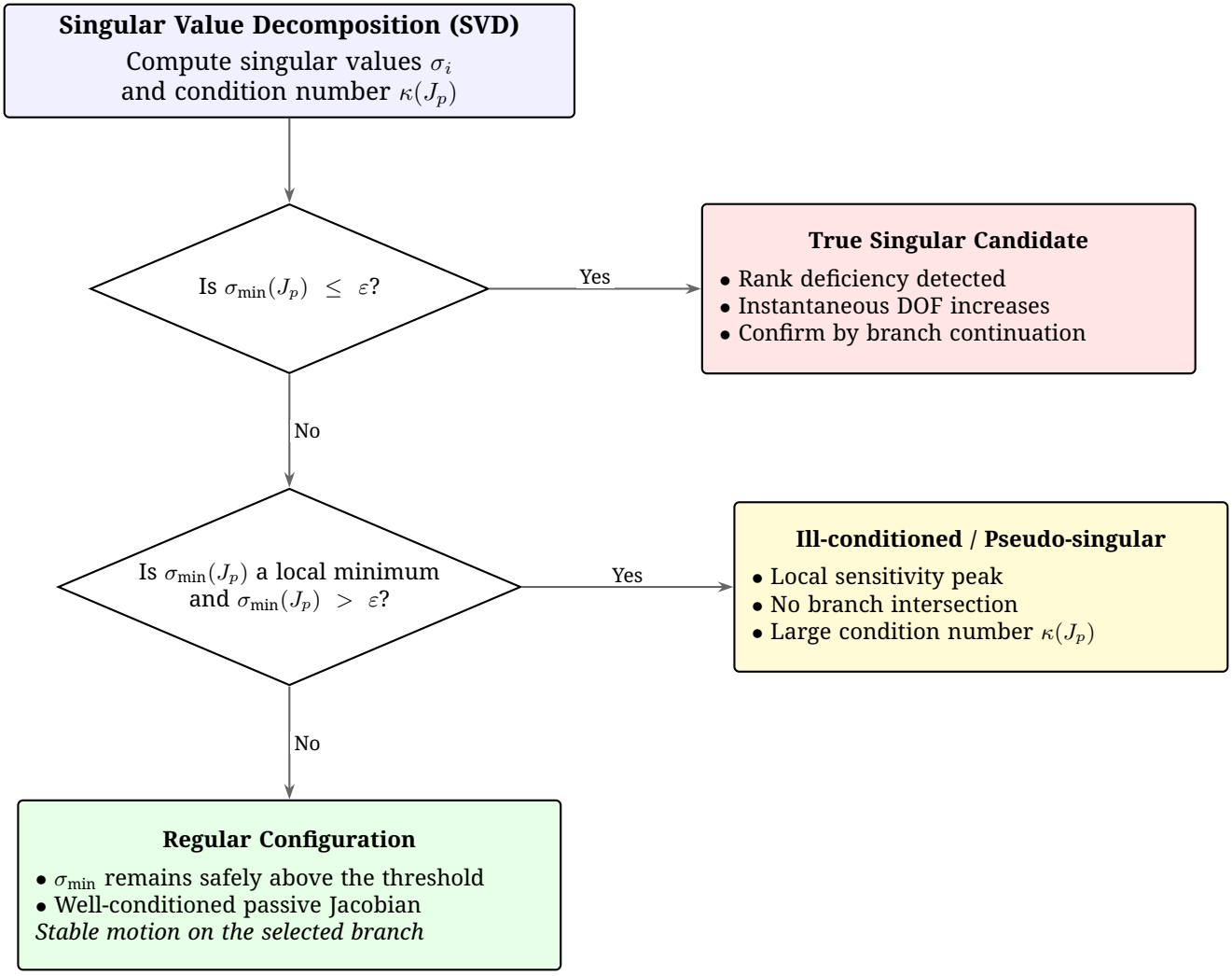

Singular Value Decomposition (SVD)
Compute singular values σ_i
and condition number $\kappa(J_p)$
Is $\sigma_{\min}(J_p) \leq \varepsilon$?
Yes
True Singular Candidate
• Rank deficiency detected
• Instantaneous DOF increases
• Confirm by branch continuation
No
Is $\sigma_{\min}(J_p)$ a local minimum
and $\sigma_{\min}(J_p) > \varepsilon$?
Yes
Ill-conditioned / Pseudo-singular
• Local sensitivity peak
• No branch intersection
• Large condition number $\kappa(J_p)$
No
Regular Configuration
• $\sigma_{\min}$ remains safely above the threshold
• Well-conditioned passive Jacobian
Stable motion on the selected branch